

Ebelechukwu Nwafor

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Education

- Howard University, Dept. of Systems and Computer Science** Washington, DC
Doctor of Philosophy in Computer Science May 2018
Dissertation: “Trace-based Data Provenance for Cyber-Physical Systems”
Advisor: Dr. Gedare Bloom
- Norfolk State University, Dept. of Computer Science** Norfolk, VA
Master of Science in Computer Science May 2013
Thesis: “Prototype Implementation of a Self-Protecting Security Framework for CDA Documents”
Advisor: Dr. George Hsieh
- Cheyney University of Pennsylvania** Cheyney, PA
Bachelor of Arts in Computer and Information Science May 2011

Appointments

- Villanova University** Villanova, PA
Associate Professor Aug 2025 – Present
- Teach courses including Introduction to Cybersecurity, Introduction to Python, Principles of Database Systems, and Security and Privacy in Cyber-Physical Systems.
 - Conduct research on Internet of Things (IoT) security; contribute to university service activities.
- Assistant Professor (Tenure-Track)* Aug 2019 – Aug 2025
- Taught core undergraduate and graduate courses in cybersecurity, Python, database systems, and cyber-physical systems security.
 - Maintained an active research agenda in IoT security; mentored students and participated in departmental service.
- Middlebury Institute of International Studies** Monterey, CA
Affiliated Faculty — Master’s in Cybersecurity Program Summer 2025
- Designed and taught the Network Security course for the Master’s in Cybersecurity program, covering core concepts in network defense, threat analysis, and secure system architecture.
- Howard University** Washington, DC
Instructor — RCMI Data Science Initiative Aug 2019 – Present
- Teach Introduction to Data Science for the Research Centers in Minority Institutions (RCMI) Data Science Initiative, supporting foundational training in data science for underrepresented researchers.
- Assistant Professor — Google TechExchange* Aug 2018 – May 2019
- Instructor for the Howard University–Google TechExchange initiative; taught Theory of Computation and Introduction to Networking to a competitive cohort of HBCU students.
- Embedded Systems and Security Lab, Howard University** Washington, DC
Research Assistant / Project Lead Jan 2016 – Jul 2018
- Led development of a provenance data collection framework for memory-constrained cyber-physical systems, enabling anomaly detection on resource-limited embedded devices.
- IBM** San Jose, CA
Software Engineering Research Intern Jun 2016 – Aug 2019

- › Migrated the search framework for the IBM LASO / Medical Sieve project from Apache Lucene to Elasticsearch; developed a Java bulk-ingestion application using the JEST client library.
- › Benchmarked Elasticsearch cluster performance against Apache Lucene and researched efficient serialization strategies for large multi-index document collections.

IBM Research-Almaden

Research Intern

San Francisco Bay Area, CA

May 2015 – Aug 2015

- › Applied information extraction techniques and user context modeling to enhance the security of enterprise systems.

Publications

1. Agbaje, P., Mitra, A., Anjum, A., Khose, P., Nwafor, E., Olufowobi, H. "Enhancing Graph Neural Networks: A Mutual Learning Approach." *arXiv preprint arXiv:2510.19223*. 2025.
2. Nwafor, E., Nguyen, M.P. "Fostering Digital Inclusion for Low-Resource Nigerian Languages: A Case Study of Igbo and Nigerian Pidgin." *Proceedings of the Eighth Workshop on Technologies for Machine Translation of Low-Resource Languages*. 2025.
3. Nwafor, E., Baskota, U., Parwez, M.S., Blackstone, J., Olufowobi, H. "Evaluating Large Language Models for Enhanced Intrusion Detection in Internet of Things Networks." *2024 IEEE Global Communications Conference (GLOBECOM)*, pp. 3358–3363. 2024.
4. Kolimago, C., Robson, M., Olufowobi, H., Nwafor, E. "Simulating Load Sharing for Resource Constrained Devices." *IEEE Access*, vol. 12, pp. 157021–157038. 2024.
5. Landry, L., Lucas, M., Andy, A., Nwafor, E. "Artificial Intelligence Assisted Curation of Population Groups in Biomedical Literature." *International Journal of Digital Curation*, vol. 18, no. 1, p. 9. 2024.
6. Anjum, A., Agbaje, P., Mitra, A., Oseghale, E., Nwafor, E., Olufowobi, H. "Towards Named Data Networking Technology: Emerging Applications, Use Cases, and Challenges for Secure Data Communication." *Future Generation Computer Systems*, vol. 151, pp. 12–31. 2024.
7. Nwafor, E., Schmidt, C., Olufowobi, H. "IoT-MGSec: Mitigating Man-in-the-Middle Attacks in IoT Networks using Graph-Based Learning." *22nd IEEE International Conference on Machine Learning and Applications (ICMLA)*. 2023.
8. Agbaje, P., Anjum, A., Mitra, A., Hounsinnou, S., Nwafor, E., Olufowobi, H. "Privacy-Preserving Intrusion Detection System for Internet of Vehicles using Split Learning." *10th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT)*. 2023.
9. Nwafor, E. "A Game-Theoretic Approach to Detecting Romance Scams." *New Perspectives in Behavioral Cyber-security*, pp. 157–159. 2023.
10. Agbaje, P., Nwafor, E., Olufowobi, H. "Deep Reinforcement Learning for Energy-Efficient Task Offloading in Cooperative Vehicular Edge Networks." *21st IEEE International Conference on Industrial Informatics (INDIN)*, pp. 1–8. 2023.
11. Agbaje, P., Anjum, A., Talukder, Z., Islam, M., Nwafor, E., Olufowobi, H. "FedCime: An Efficient Federated Learning Approach for Clients in Mobile Edge Computing." *IEEE International Conference on Edge Computing and Communications (EDGE)*. 2023.
12. Nwafor, E., Olufowobi, H. "CANBERT: A Language-Based Intrusion Detection Model for In-Vehicle Networks." *21st IEEE International Conference on Machine Learning and Applications (ICMLA)*. 2022.
13. Nwafor, E., Andy, A. "A Survey of Machine Translation Tasks on Nigerian Languages." *Proceedings of the 13th International Conference on Language Resources and Evaluation (LREC)*. 2022.

🔗 Full publication list (17 papers): scholar.google.com/citations?user=zoHDXwEAAAAJ